SectraTEK

AI FOR CYBERSECURITY AND DIGITAL FORENSICS · LLMS FOR FORENSICS

# Ask the Bug Report



A forensic chatbot over an Android bug report: ask in plain English, and get an answer that cites the section it came from. Use a report from your own phone.

Daniel Kwaku Ntiamoah Addai · Mobile Threat Research Lead, SectraTEK Lab · AI for Cybersecurity and Digital Forensics

THE IDEA

# From a whole extraction to a single text file

A full phone extraction is large and needs special tooling. An Android bug report is just a text file, so any student can make one and practise asking a model grounded, cited forensic questions.

A FULL EXTRACTION

### Gigabytes, special tools

Powerful, but heavy to acquire and store. Hard to hand to every student.



A BUG REPORT

### One sectioned text file

Lightweight, and you can pull one from your own phone with a single command.

THE REPORT

# What is inside an adb bug report

The output of `adb bugreport` is one text file split into labelled sections. The ones that matter here:

### Device

Model, Android version, build - the device fingerprint.

### Packages

The apps installed on the device.

### Accounts

The accounts configured on the device.

### Wi-Fi

Saved networks and connection state.

### Battery

Charge level and power state.

### System log

Recent events, errors, and crashes.

THE PROBLEM

# Ask a model directly, and it invents

> *"This phone has Gmail, WhatsApp, and a banking app installed."*

WHY THAT IS A PROBLEM

### It never saw the report

The model answers from general training, not this device.

### It invents specifics

Plausible app names and settings that are not in the file.

### Nothing to check

No section is named, so the answer cannot be verified.

The fix is to give the model the report and read-only tools, so it answers from the file and cites it.

THE AGENT

# Three read-only tools over the report

The model never reads the whole file blindly. It is given three tools and decides which to call.



```
list_sections()
```

List the report's sections, so the agent knows what is available.



```
read_section(name)
```

Read one section, by number or by text in its title.



```
search_report(keyword)
```

Search the whole report; return matching lines with their section.

Read-only: the agent can list, read, and search, but never change the report.

HOW IT WORKS

# Think, act, observe - then answer with the section

### Question

A plain-English forensic question.

### Think + act

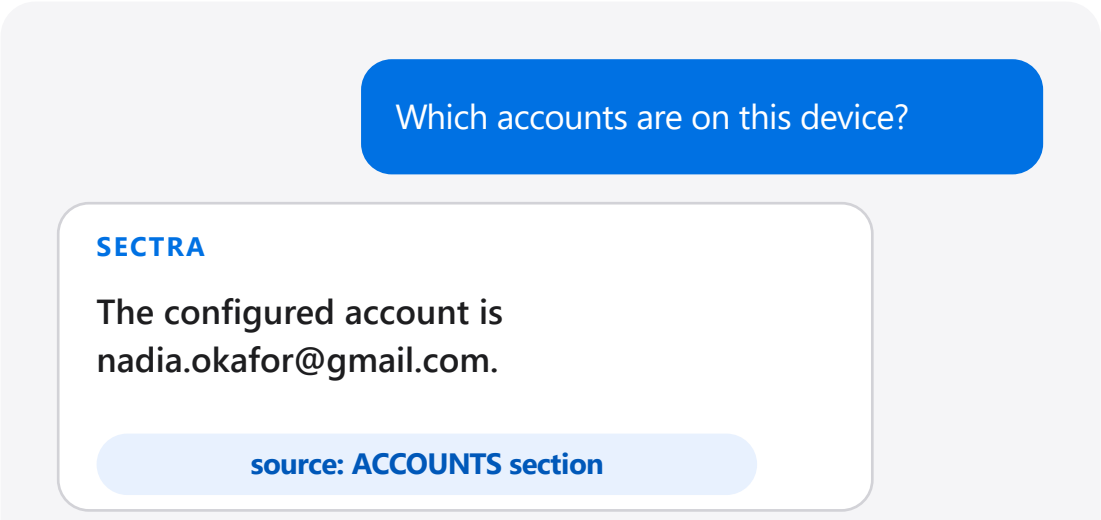
Pick one tool and call it.

### Observe

Read what the tool returned.

### Answer

Reply, and cite the section.




### It loops until it can answer

Search, read, repeat - one tool per step.

### It cites the section

Every answer names where it came from.

### You can verify it

Open that section and check for yourself.

GET ONE

# Pull a bug report from your own phone

| | | |
|---|---|---|
| 1 | **Enable Developer Options** | On the phone, tap Build number seven times, then turn on USB debugging. |
| 2 | **Connect and run adb** | On a computer with the Android platform tools, connect the phone. |
| 3 | **Produce the file** | adb bugreport makes a .zip (or .txt). Upload it into the chatbot. |
| 4 | **Or use the sample** | A synthetic sample_bugreport.txt is bundled, so it works before you have your own. |

```
adb bugreport  mybugreport      # produces mybugreport.zip - then upload it
```

YOUR TURN

# Open the chatbot and ask it something

| | | |
|---|---|---|
| 1 | **Open in Colab** | Click the badge in bugreport_chatbot_ui.ipynb. Nothing to install. |
| 2 | **Paste a free key** | A free OpenRouter key, or switch to local Ollama for no key. |
| 3 | **Load a report** | Use the bundled sample, or upload your own adb bugreport. |
| 4 | **Ask away** | What phone is this? Which apps? Any errors in the log? |
| 5 | **Check the source** | Read the answer, then open the cited section and confirm it. |

SectraTEK

# Ask the report. Check the section.

Same agent idea as the device-image lab, pointed at a bug report you can pull from your own phone - and every answer still cites its source.

